

Shayan Shakeri

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PROFESSIONAL SUMMARY

AI Research Engineer working on LLM agents at BMO's AI Centre of Excellence; co-founder of Merlyn Labs, a 3-person research collective. Built the deterministic eval pipeline that caught a GenAI tool serving \$200B+ AUM downplaying investment risk; now building a graph-based agentic system over bank client data and the infrastructure that provisions LLM agents from a written role and scope definition. Open-sourced a flow-matching VLA integration for RLinf; M.Eng in AI & Robotics, University of Toronto, expected April 2027.

CORE COMPETENCIES

Agentic Systems

Agent Evaluation

LLM Fine-tuning

Reinforcement Learning

Python & C/C++

PyTorch & JAX

WORK EXPERIENCE

BMO AI Centre of Excellence Sep 2025 - Present

AI Research Engineer

Toronto, Canada

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving \$200B+ AUM wealth management.
- Built a deterministic agent eval pipeline using hundreds of synthesized inputs to detect misaligned outputs at scale.
- Building a graph-based agentic system answering complex multi-hop relational queries across bank client data.
- Building internal infrastructure that provisions LLM agents from a written role and scope definition.
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy.

Merlyn Labs Aug 2025 - Present

AI Researcher & Co-Founder

Toronto, Canada

- Open-sourced a flow-matching VLA integration for RLinf, enabling RL training on the BEHAVIOR-1K suite in OmniGibson.
- Developing VLM judges that score rollouts into dense, context-dependent RL rewards that are difficult to game.
- Showed $\pi 0.5$ generalization failures on LIBERO-PRO are recipe-induced overfitting; proposed an alternative baseline.
- Conservative finetuning doubled position-swap success (21% to 42%), showing spatial generalization over memorization.

Epineuron Technologies May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op

Mississauga, Canada

- Helped develop PeriPulse, an FDA Breakthrough-designated device now in multi-national clinical trials; designed and assembled PCBs and authored IEC 60601-1 / ISO 13485 validation protocols.
- Optimized power draw via oscilloscope and power-analyzer testing, achieving 900% battery life improvement.

PROJECTS

Stanford BEHAVIOR-1K Challenge 8th Place, Standard Track

Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); found chunked execution outperformed temporal ensembling 3x. Trained on 10,000+ demonstrations covering 22 of 50 scored task types; published a technical report and a LessWrong analysis.

BardSong Closed Alpha

Built the end-to-end pipeline: Groq speech-to-text, Gemini scene extraction, image generation, narrated video. Fine-tuned an LLM on D&D sourcebooks for narrative coherence; in closed alpha with 23 dungeon masters.

EDUCATION

M.Eng in AI & Robotics University of Toronto Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics Co-op McMaster University Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Agents & ML: Agents, Agent Evaluation, Alignment Evaluation, LLM Fine-tuning, Reinforcement Learning, Imitation Learning, Flow Matching
Systems & Languages: Python, C/C++, PyTorch, JAX, OmniGibson, MuJoCo, ROS2, Sim-to-Real, PCB Design